



Thermal failure analysis in semiconductor devices through integrated multiphysics measurement and modeling

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ABSTRACT

The performance of semiconductor devices is strongly governed by multi-physics coupling effects. Accurately characterizing the internal distributions of multiple physical fields and performing coupled modeling analysis are critical for investigating thermal failure mechanisms and guiding design optimization. However, the lack of integrated measurement techniques capable of simultaneously capturing multiple physical quantities has made comprehensive device evaluation challenging. To overcome this limitation, this study develops an integrated measurement system that enables multi-parameter characterization in semiconductor devices, including temperature rise, residual stress, thermal stress, carrier mobility, carrier concentration, and bandgap. Using a GaN high-electron-mobility transistor (HEMT) as a test vehicle, we measured the spatial distributions of multiple physical parameters along the channel length direction and established a coupled model that accounts for the interplay among thermal, mechanical, and electrical properties. Experimental and modeling analyses reveal that localized tensile thermal stress at the device edges serves as a primary driver of failure, offering important insights for future comprehensive performance optimization of semiconductor devices.

1. Introduction

Semiconductor devices are critical for converting and controlling electrical energy, with applications spanning from micro-electronics to industrial power systems [1]. With the increasing demand for performance, these devices are operating at higher power densities and frequencies, which inevitably leads to cooling problem [2–4]. The resulting high temperature not only affects device efficiency but also significantly reduces the lifespan of semiconductor materials [5,6]. Hot spots also lead to thermal stress, which can induce mechanical failures and further degrade device reliability [7]. Cooling problem has become a major cause of semiconductor device failure, however, it should be noted that the devices' failure mechanism is not determined by a single factor, but rather by the complex interactions between multiple coupled physical fields [8,9]. There are obvious effects of temperature on the thermal or electrical properties of materials [10–12], and stress also has a significant effect on the material properties. For example, under certain tensile stress, the room temperature thermal conductivity of gallium nitride (GaN) can be reduced obviously [13,14]. Stress can also greatly change the bandgap and carrier mobility of the materials [15,16]. Therefore, in

the thermal design of semiconductor devices, focusing solely on "thermal factors" is no longer sufficient. It is essential to reveal the principles of multiphysics interactions in order to accurately analyze failure mechanisms and achieve further optimization.

Numerous simulation studies have been devoted to analyzing the coupling effects of thermal, mechanical, and electrical multiphysics fields [17–19]. However, a major challenge remains the extreme scarcity of real distribution data for these internal multiphysics fields in operating semiconductor devices, which often limits simulation results to qualitative guidance rather than quantitative analysis. Experimental research by Li et al. [20] reveals a strong spatial correlation between the failure locations in power devices and regions of electric field concentration. This indicates that device failure is not solely caused by thermal effects alone, but is likely the result of thermoelectric multiphysics coupling leading to stress generation, which subsequently affects thermal and electrical performance, ultimately causing device failure. Nevertheless, their study has not yet achieved direct measurement of the stress distribution within the device, and further investigation is required. Therefore, achieving in situ measurement of multiphysics fields and key parameters inside power devices has become a crucial prerequisite for optimizing multiphysics-coupled designs.

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Nomenclature			
A_i	intrinsic temperature - Raman shift coefficient [$\text{cm}^{-1}\text{K}^{-1}$]	n_ω	Bose-Einstein factor
A_m	effective temperature - Raman shift coefficient [$\text{cm}^{-1}\text{K}^{-1}$]	ε_ω	dielectric function
c	speed of light [m/s]	ε_∞	high frequency dielectric function
e	electron charge [C]	γ	Damping constant of plasmon [$1/\text{cm}$]
E	Young's modulus [Nm^{-2}]	Γ	Damping constant of plasmon-couple phonon mode [$1/\text{cm}$]
E_g	bandgap energy [eV]	ν	Poisson's ratio
f_p	constraint factor, proportional coefficient of stress variation and temperature rise [$\text{Nm}^{-2}\text{K}^{-1}$]	μ	carrier mobility [$\text{cm}^2\text{V}\cdot\text{s}$]
h	Planck's constant [J·s]	λ_{PL}	wavelength of the main photoluminescence peak [m]
I_A	Raman spectral intensity	Subscripts	
I_{PL}	photoluminescence spectral intensity	*	effective value
I_0	basic intensity fitting parameter	f	film layer
k_B	Boltzmann constant [J/K]	s	substrate layer
K	intrinsic stress - Raman shift coefficient [$\text{cm}^{-1}\text{N}^{-1}\text{m}^2$]	t	$A_1(\text{TO})$ mode
m^*	effective electron mass [kg]	l	$A_1(\text{LO})$ mode
n	carrier concentration [cm^{-3}]	p	plasmon mode
T	thermodynamic temperature [K]	PL	photoluminescence mode
V_{ds}	drain-to-source voltage [V]	Acronyms	
V_{gs}	gate-to-source voltage [V]	CCD	Charge-coupled device
x	x-axis	CW	Continuous-wave
Greek symbols		DC	Direct current
α	coefficient of thermal expansion [K^{-1}]	DIC	Digital image correlation
ΔT	temperature rise [K]	HEMT	High electron mobility transistor
$\Delta\sigma$	thermal stress variation [N/m^2]	IR	Infrared
$\Delta\omega$	Raman shift [$1/\text{cm}$]	IGBT	Insulated gate bipolar transistor
ξ	correction factor	PCB	Printed circuit board
σ	stress	PL	Photoluminescence
ω	spectral frequency [$1/\text{cm}$]	TEM	Transmission electron microscopy
ω_p	plasmon frequency [$1/\text{cm}$]	TOF	Time-of-flight
		XRD	X-ray diffraction

There have been many methods for measuring the key physical quantities of semiconductor devices. For temperature measurement, common electrical methods include thermocouple method, thermal resistance method, and other methods based on temperature-electrical changed characteristics [21], while optical methods include infrared (IR) spectroscopy [22] and thermos-reflectance imaging [23]. For stress measurement, there are digital image correlation (DIC), transmission electron microscopy (TEM), and X-ray diffraction (XRD) [24]. For electrical properties measurement, methods such as the Hall effect method [25], time-of-flight (TOF) method [26], optical absorption spectroscopy [27] can measure the key electrical properties of semiconductor materials, including carrier mobility and bandgap. It can be found that, among commonly used measurement methods, there are significant differences in the measurement principles of different physical quantities. If various measuring instruments are just integrated directly, it will lead to complex hardware design and cumbersome operation. It is prone to mutual interference between different methods and reduce the accuracy of the measurement as literatures [28]. Meanwhile, the spatial resolution of most methods is not high enough, and they can only measure the differences between multiple samples. It is difficult to measure the distribution of multiple physical quantities of a single sample. For example, the size of the X-ray light source of the XRD method is more than a few hundred microns, while the spatial resolution of the Hall method can be even lower because it is based on electrical probes. To achieve high-precision comprehensive measurement, there is an urgent need to develop a comprehensive strategy that can integrate similar sensors, enable detection without interference, and unify measurement principles as much as possible.

Raman spectroscopy is expected to be a unified measurement platform for the comprehensive measurement. It is sensitive to temperature,

stress and even electrical field, which also offers high precision, strong penetration capability, material identification capability, and submicron spatial resolution, providing significant advantages in the measurement of semiconductor devices [29–31]. Substantial research has been dedicated to the decoupling of multiple physical fields using Raman detection. Graham et al. decoupled the effects of temperature and stress utilizing the full width at half maximum (FWHM) and peak position of a Raman band [32,33]. Subsequently, Batten et al. [34] and Choi et al. [35] independently demonstrated simultaneous temperature and stress measurement in semiconductor devices by employing two characteristic Raman peaks. Building on this foundation, Wang et al. [36] further advanced the technique by accounting for the influence of electric fields on Raman peak positions. They proposed a linear three-peak decoupling method based on the shifts of three Raman features, enabling the simultaneous decoupling of temperature, stress, and electric fields. In 2023, addressing the challenge posed by semiconductors possessing only a single Raman peak with a high signal-to-noise ratio, our group introduced a two-step Raman method [37]. This approach enables temperature-stress decoupling based solely on the peak position of a single Raman band by comparing the equivalent Raman-temperature shift coefficient with the intrinsic one for calibration. It is worth noting that, Raman spectroscopy is not only widely used in temperature or stress measurement [21,38], but also can obtain the carrier concentration and mobility of semiconductor materials by analyzing plasmon-coupled phonon modes [39]. In addition, based on the same grating and charge-coupled device, Raman spectrometer can also obtain bandgap by measuring photoluminescence (PL) spectra [40].

The aforementioned non-contact measurement methods lay the foundation for in-situ and comprehensive measurement of multiple influencing factors within power devices, making it possible to establish

realistic multi-physics coupling models and reveal device failure mechanisms. The successful integration of UV-Vis Raman measurements in previous work [41,42] further demonstrates the feasibility of combining multiple measurement modalities. Building on this research, this paper proposes an integrated analysis strategy that combines previously independent measurement techniques into a unified system, enabling high-spatial-resolution detection of temperature, stress distribution, carrier mobility, carrier concentration, and bandgap in semiconductor devices. Based on this integrated Raman-photoluminescence (Raman-PL) detection system, the distributions of residual stress, structural constraint factors (representing the thermal stress generated per unit temperature rise), carrier mobility and concentration, and bandgap along the channel of a GaN high electron mobility transistor (HEMT) device were determined. A multi-physics coupling model was established based on the measured data. By comparing theoretical predictions with experimental measurements, it was found that the most vulnerable region of the semiconductor device is not the point of maximum temperature rise in the central area at low voltage, but rather the tensile stress concentration points at the edge regions. Thermal stress was further shown to trigger significant changes in carrier mobility, bandgap, and thermal conductivity, creating a vicious cycle under high-temperature conditions. This insight provides a new direction for the comprehensive optimization of future semiconductor devices.

2. Analytical model and measurement method

2.1. Analytical model

Using temperature as a benchmark for evaluating the heat dissipation performance of semiconductor devices. The temperature distribution of semiconductor devices can be described by the heat conduction equation with an internal heat source, assuming that the temperature rise of the semiconductor layer of the power device is θ , then it should be described by Eq. (1):

$$\frac{d^2\theta}{dx^2} + \frac{d^2\theta}{dy^2} + \frac{\Phi}{\lambda} = 0 \quad (1)$$

Where, Φ is the internal heat source generated by the power device. And it relates to the gate source voltage, drain source voltage, carrier mobility, dielectric constant, carrier concentration and other parameters.

For a GaN-based HEMT, the equivalent resistance is inversely proportional to the carrier concentration and carrier mobility, while the source-drain current is directly proportional to the dielectric constant and carrier mobility.

$$\Phi = i_{ds}^2 R_{on} \quad (2)$$

Where, R_{on} can be expressed as Eq. (3) [43]:

$$R_{on} = \frac{L_{ds}^2}{q\mu_{2D}n_s} \quad (3)$$

Where L_{ds} is the distance between drain and source, q is elementary charge, the sheet carrier density on the 2D electron gas, n_s , may be determined by means of Hall bar measurements and/or capacitance measurements. μ_{2D} is in this case the mobility of electrons in the 2DEG channel. The sheet carrier density on the 2D electron gas, n_s .

The current i_{ds} can be expressed as:

$$i_{ds} = \beta \left(V_{gs} - V_t - \frac{1}{2} V_{ds} \right) V_{ds} \quad (4)$$

Where,

$$\beta(\sigma, T) = \mu_{2D}(\sigma, T) \epsilon_{ox}(\sigma, T) W/L \quad (5)$$

Therefore, with a certain structure, and the certain drain to source

voltage and gate to source voltage, the heat source of HEMT can be described as:

$$\Phi = k_1 \frac{\mu_{2D} \epsilon_{ox}^2}{n_s} \quad (6)$$

Where k_1 is a constant related to the structure characteristics.

Therefore, the Eq. (1) can be rewritten as:

$$\frac{d^2\theta}{dx^2} + \frac{d^2\theta}{dy^2} + \frac{k\epsilon_{ox}^2\mu_{2D}}{\lambda n_s} = 0 \quad (7)$$

Where ϵ_{ox} is dielectric constant, μ_{2D} is carrier mobility of two-dimensional electron gas (2DEG), λ is thermal conductivity, k is a constant depended on the power device. Meanwhile, ϵ_{ox} , μ_{2D} , and λ are function of stress σ , and temperature T . And stress σ can be further divided into the sum of residual stress σ_0 and thermal stress $\sigma_T = f_p\theta$, where f_p is a parameter reflecting the constraint that cause thermal stress, and which is defined as the structure constraint factor. With an optimized interface, μ_{2D} is dictated by the intrinsic mobility of the GaN channel. Therefore, stress-induced changes to the intrinsic GaN mobility, μ , will correspondingly affect the 2DEG transport properties, thus, the μ_{2D} can be approximated as a function of μ with a coefficient of k_2 . And the heat conduction equation can be described as:

$$\frac{d^2\theta}{dx^2} + \frac{d^2\theta}{dy^2} + kQ_d(\sigma_0, f_p, \theta) = 0 \quad (8)$$

Where k is related to k_1 and k_2 , $Q_d = \epsilon_{ox}^2\mu/\lambda n_s$, it can determine the temperature distribution of power device, and it is a function of residual stress σ_0 , constraint factor f_p , and temperature rise θ . And the f_p can be determined by Raman calibration. Integrally heating the target device, the effective temperature - Raman shift coefficient A_m at different position (x, y) can be calibrated, and the constraint factor f_p is expressed as Eq. (9).

$$f_p(x, y) = \frac{A_m(x, y) - A_i}{K} \quad (9)$$

Where $\Delta\omega$ is the Raman shift, ΔT is the temperature rise, K is the instinct stress-Raman shift coefficient, and A_i is the instinct temperature-Raman shift coefficient. Meanwhile, there is a correction coefficient ξ for correcting the temperature rise difference caused by the uneven distribution of the upper and lower layers can be calculated based on the following expression [37]:

$$\xi = \frac{E \left(\alpha_s \frac{\Delta T_s}{\Delta T_f} - 2\alpha_f \right)}{f_p(1 - \nu)} + 1 \quad (10)$$

where E is the Young's modulus of the material, ν is Poisson's ratio, ΔT_f and ΔT_s are the pre-measured temperature rises of the film layer and the substrate layer, respectively, and α_f and α_s are the thermal expansion coefficients of the film layer and the substrate layer, respectively. Then the measured temperature rise ΔT and thermal stress $\Delta\sigma$ can be given by the following two expressions [37]:

$$\Delta T = \frac{\Delta\omega}{\xi A_m + [1 - \xi] A_i} \quad (11)$$

$$\Delta\sigma = \frac{\Delta\omega \xi [A_m - A_i]}{[\xi A_m + [1 - \xi] A_i] K} \quad (12)$$

Raman measurement of carrier mobility and concentration requires analysis of the plasmon-phonon coupling modes. Taking GaN as an example, the $A_1(\text{LO})$ coupling mode can be fitted by the following expression [44]:

$$I_A = I_0 \cdot (n_{\omega} + 1) \cdot A \cdot \text{Im} \left(-\frac{1}{\epsilon_{\omega}} \right) \quad (13)$$

where I_A is the Raman spectral intensity, I_0 is the basic fitting parameter, n_ω is the Bose-Einstein factor, and ε_ω is the dielectric function. The Bose-Einstein factor n_ω is related to spectral frequency ω and thermodynamic temperature T , and A is a function related to deformation potential scattering and electro-optical scattering mechanisms. More details can be found in Supporting Information. The dielectric function is expressed as [39]:

$$\varepsilon_\omega(\omega) = \varepsilon_\infty \left(1 + \frac{\omega_l^2 - \omega_t^2}{\omega_t^2 - \omega^2 - i\omega\Gamma} - \frac{\omega_p^2}{\omega^2 + i\omega\gamma} \right) \quad (14)$$

where ε_∞ is the high frequency dielectric constant, ω_l and ω_t are the frequencies of $A_1(\text{LO})$ and $A_1(\text{TO})$ modes respectively, Γ is the damping constant of plasmon-coupled phonon mode ($A_1(\text{LO})$ or $A_1(\text{TO})$), and ω_p and γ are the plasmon frequency and damping constant respectively. After fitting ω_p and γ , carrier concentration n and mobility μ can be obtained by substituting them into the following two expressions [45]:

$$n = \frac{\omega_p^2 m^* \varepsilon_\infty}{4\pi e^2} \quad (15)$$

$$\mu = \frac{ce}{m^* \gamma} \quad (16)$$

where m^* is the effective electron mass, e is the electron charge, and c is the speed of light.

The Raman spectrometer can be used to excite the material with energy greater than the material bandgap width and collect the PL spectrum of the material. Then, the position of the main peak of the PL spectrum can be determined to obtain the bandgap E_g of the material [40]:

$$E_g = \frac{hc}{\lambda_{PL}} \quad (17)$$

where h is Planck's constant and λ_{PL} is the wavelength of the main peak

of the PL spectrum. If a higher-precision bandgap measurement is required, the vertical coordinate of the spectrometer measurement window can be adjusted to nm units, and the PL peak curve can be fitted by referring the following theoretical expression [46]:

$$I_{PL} = I_0 \sqrt{E - E_g} \exp\left(-\frac{(E - E_g)e}{k_B T}\right) \quad (18)$$

where I_{PL} is PL spectral intensity, E is energy value, and k_B is the Boltzmann constant. Of course, to simplify the processing, the PL spectrum of the material can be directly separated into multiple peaks and the bandgap can be determined by fitting the position of the main peak (as shown in Supporting Information).

2.2. Integrated measurement system

Combining the Raman decoupling temperature stress method we proposed previously, the existing plasmon-coupled phonon fitting method and photoluminescence spectroscopy method, this work integrated a comprehensive spectral measurement method. This integrated method can not only measure the residual stress distribution of the device, analyze the carrier mobility, concentration and bandgap of the device material, but also decouple the temperature rise and thermal stress distribution during the device operation. The integrated measurement system is shown in Fig. 1(a). The system is mainly composed of PL laser, Raman laser, edge filter, lens, beam splitter, grating, objective, charge coupled device (CCD), and computer processing program. The lasers are all continuous wave (CW) polarized. Since PL spectroscopy measurement requires lasers with wavelength energy greater than the material bandgap value, PL spectroscopy measurement may require additional short-wavelength lasers, such as lasers with wavelengths less than 1240/3.4 $\approx$ 365 nm for GaN measurement. Raman spectroscopy uses a longer wavelength laser, which can improve the spatial resolution of the measurement on the one hand and avoid the interference of fluorescence on the other hand. Of course, in the case of not high

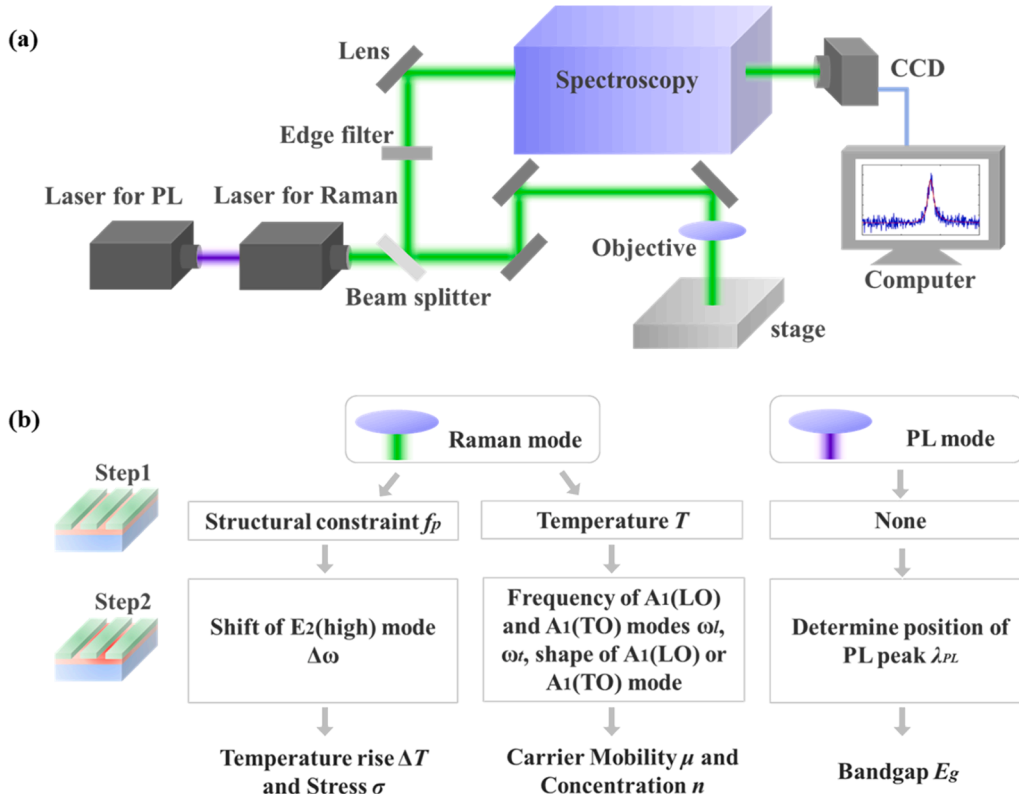


Fig. 1. (a) Integrated spectroscopy measurement system and (b) comprehensive measurement steps of integrated method for GaN.

accuracy requirement, the same wavelength laser can also be used to measure the two kinds of spectra at the same time. In addition, for most semiconductor materials, the PL spectrum and Raman spectrum are often not in the same optical window. And the spectral width of the two is significantly different, the Raman spectrum is narrower ($\sim 10 \text{ cm}^{-1}$), and the PL spectrum is wider ($\sim 1000 \text{ cm}^{-1}$). Therefore, the optical window needs to be adjusted to the corresponding spectral range. Since switching the optical window will bring non-negligible systematic errors (mainly affecting the Raman spectrum with higher accuracy requirement), the system should not switch to the optical window to measure another kind of spectrum without completing the current spectrum measurement.

The comprehensive measurement steps of the integrated method are shown in Fig. 1(b). The system includes two modes, Raman mode and PL mode. The temperature and thermal stress can be decoupled by determining the peak position of a single peak in Raman mode, and the carrier concentration and mobility can also be obtained by analyzing the plasmon-coupled phonon in Raman mode. The photoluminescence mode can obtain the bandgap by determining the main peak position of PL spectra curve. Take GaN as an example: Since $E_2(\text{high})$ Raman mode of GaN has the highest signal-to-noise ratio, this peak was chosen to measure temperature and stress. GaN has two plasmon-coupled phonon modes, $A_1(\text{TO})$ is a low frequency mode, $A_1(\text{LO})$ is a high frequency mode. Therefore, both phonon modes can be used to analyze carrier physical properties. The high frequency mode $A_1(\text{LO})$ has a significantly higher signal-to-noise ratio, so it was chosen for subsequent analysis in this work. The PL spectrum of GaN is a complex curve with multiple peaks superimposed together, and the position of the main peak is about 363–365 nm. Other semiconductor materials can also achieve comprehensive measurement of these physical quantities by selecting appropriate peaks in sequence like GaN [47]. In addition, the measurement of different physical quantities requires different supplementary information. For example, the decoupling measurement of temperature and stress requires the determination of structural constraint factors first, and the analysis of carrier physical properties requires the thermodynamic temperature of the material at that time. The steps for measuring different physical quantities are outlined in Fig. 1(b).

3. Measurement and results

The sample measured in this work was a GaN HEMT device (Cree CGH60008D). The image of the device under optical microscope is shown in Fig. 2(a). The device has a total of 6 channels, and the two ends of the length direction of the channel correspond to two electrodes loaded with Gate voltage and Drain voltage respectively. Assume that the length direction is the x direction, and the x origin is set as the

junction between the Gate side and the middle area. The length of the device from the origin to the junction between the Drain side and the middle area is about 360 μm . A confocal micro-Raman spectrometer (HORIBA LaRAM HR Evolution) was used for measurement, with a 600 grooves/mm grating and a $50\times$ objective. For Raman spectroscopy, a 633 nm solid-state CW laser was selected, and the integration time of a single Raman peak was 3 s. A Raman spectrum of the device in the frequency range of 50–1050 cm^{-1} is shown in Fig. 2(b). Since the substrate of the device is silicon carbon (SiC), the typical Raman peaks of the GaN layer and the SiC layer can be measured simultaneously. The $E_2(\text{high})$ peak of GaN (at about 568 cm^{-1}) and the $E_2(\text{high})$ peak of SiC (at about 778 cm^{-1}) are both used for stress measurement. In addition, the $A_1(\text{TO})$ peak of GaN (at about 538 cm^{-1}) and the $A_1(\text{LO})$ peak (at about 726 cm^{-1}) are used to analyze carrier properties.

First, the residual stress distribution of the device in the non-working state is determined. Since the in-plane size of the device is much larger than the normal size, the stress it is subjected to is mainly in-plane biaxial stress. Therefore, by scanning the Raman peak position of the device in the channel length direction, the peak position distribution can be converted into residual stress distribution through the in-plane stress-Raman shift coefficient. We selected a total of 14 points in the length range of -10–380 microns for measurement, measured the Raman spectrum 5 times at each point, and took the average value of the measured values. The ranges of -10–0 μm and 360–380 μm represent the edge area of the device on the Gate side and the edge area on the Drain side, respectively. Since the measured value has a large mutation in the edge area, we measured several more positions in the edge area. We assume that the stress values of both at $x=380\mu\text{m}$ are 0GPa, and the residual stress distribution of the GaN layer and SiC layer can be obtained as shown in Fig. 3(b). The data show that the GaN layer is subjected to an average tensile stress of about 0.2GPa (positive values indicate tensile stress, negative values indicate compressive stress) compared to the point at $x=380\mu\text{m}$, and the maximum tensile stress is 0.27GPa. The residual stress of the GaN layer has a significant mutation on the Drain side, but the change is not obvious on the Gate side. However, the SiC layer is subjected to an average tensile stress of about 0.4GPa compared to the point at $x=380\mu\text{m}$, and the maximum tensile stress is 0.57GPa. The residual stress of the SiC layer has a significant mutation at the edges on both sides. The stress non-uniformity of the SiC layer is higher than that of the GaN layer.

Then we put the device into the temperature control platform (Linkam THMS350EV-4), and uniformly changed the temperature of the device from 30°C to 230°C, and measured a set of Raman peak position distributions every 40°C rise. After sorting and fitting, the effective temperature-Raman shift coefficient temperature distribution of GaN layer and SiC layer can be obtained. The structural constraint factor distribution of GaN layer and SiC layer can be further obtained by

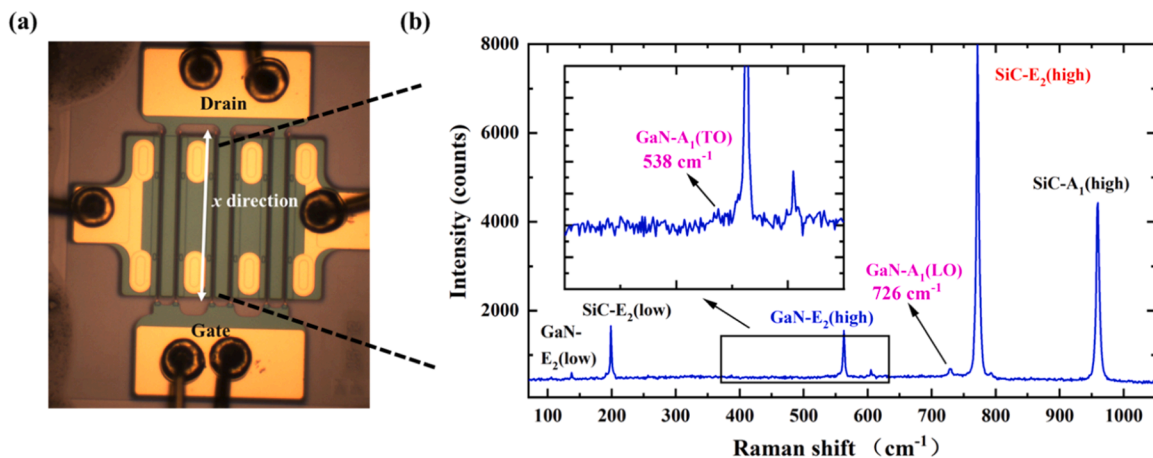


Fig. 2. (a) Optical image of CGH80008D GaN HEMT device and (b) typical Raman spectrum of the device.

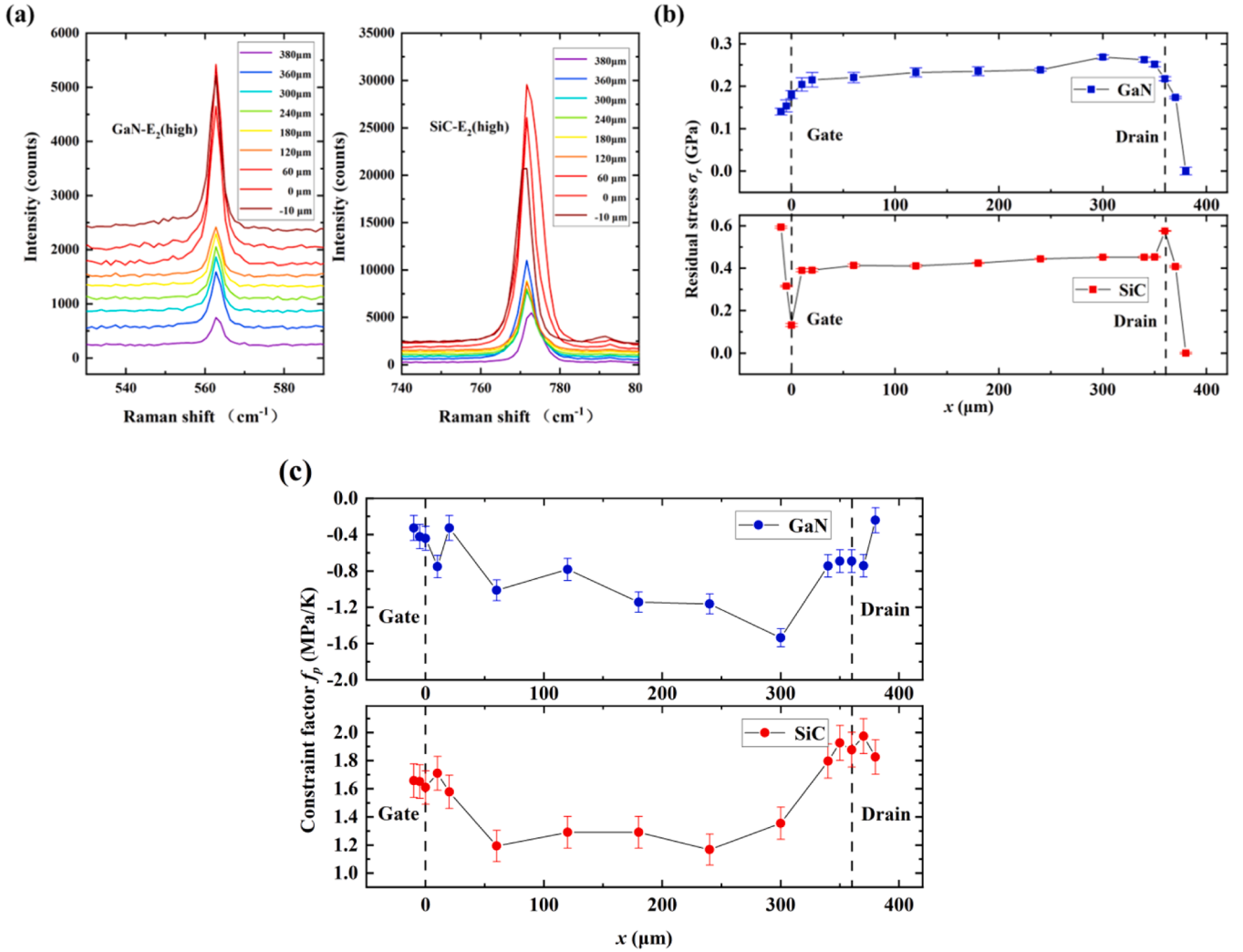


Fig. 3. (a) Raman shift distribution and (b) Residual stress distribution and (c) structural constraint factor distribution of GaN layer and SiC layer along x direction in non-working state.

expression 2, as shown in Fig. 3(c). The structural constraint factor reflects the internal constraints of the device and indicates the thermal stress generated at this position point per unit temperature rise (positive is tensile stress, negative is compressive stress). Therefore, it can be seen from Fig. 3(b) that as the temperature rises, the GaN layer and SiC layer will be subjected to tensile thermal stress and compressive thermal stress, respectively. For every 1K increase in the GaN layer, the maximum compressive stress it will be subjected to is 1.54MPa, and the minimum compressive stress is 0.24MPa. For every 1K increase in the SiC layer, the maximum tensile stress it will be subjected to is 1.97MPa, and the minimum tensile stress is 1.19MPa. However, one thing in common between the GaN layer and the SiC layer is that both ends of the two layers are subjected to tensile thermal stress compared to the central area. Tensile stress will significantly degrade the thermal and electrical properties of the material, which indicates that thermal stress will cause a certain risk of failure of the device at both ends. Although thermal stress will offset the original residual stress of the device to a certain extent, as the temperature increases, the proportion of thermal stress will gradually increase, causing its impact to gradually increase.

Based on the same set of Raman spectroscopy data for measuring the residual stress distribution of the device in the channel length direction, the carrier mobility and concentration distribution of the GaN material can also be determined. The determination of carrier properties via Raman spectroscopy relies on the analysis of longitudinal optical

phonon-plasmon coupled (LOPC) modes, with the fitting accuracy being governed by the intensity and signal-to-noise ratio of these characteristic peaks, which are fortunately well-established in semiconductors like SiC and GaN. We use expressions (6) and (7) to fit the $A_1(\text{LO})$ peak in the GaN Raman spectra, and then use expressions (8) and (9) to calculate the corresponding carrier property values. Fig. 4 (a) and Fig. 4(b) are the comparison of the experimental measurement peak and the fitting peak at the position $x = 0\mu\text{m}$ and the position $x = 180\mu\text{m}$, respectively. The formula can fit the asymmetric peak shape well. The average goodness of fit for each position point is about 0.94 or more. Through batch sorting, the carrier mobility and concentration distribution of GaN can be obtained as shown in Fig. 4 (b). The carrier mobility is smaller at the two edge areas and larger in the middle area. The minimum value is $136.8\text{cm}^2\text{V}^{-1}\text{s}^{-1}$ and the maximum value is $591.8\text{cm}^2\text{V}^{-1}\text{s}^{-1}$. The carrier concentration is larger at the two edge areas and smaller in the middle area. The minimum value is $2.47 \times 10^{17}\text{cm}^{-3}$ and the maximum value is $1.56 \times 10^{18}\text{cm}^{-3}$. While the optical approach exhibits a higher measurement uncertainty (e.g., $\sim 20\%$ for carrier mobility, about quadruple the $\sim 5\%$ uncertainty of Hall measurements), it provides a fundamentally different capability: it allows for the visualization of spatial distributions across a device, moving beyond a single, lumped-value measurement. And although the measurement accuracy of carrier mobility and concentration are affected by the signal-to-noise ratio of $A_1(\text{LO})$ peak, it can be used for trend analysis. This distribution reflects

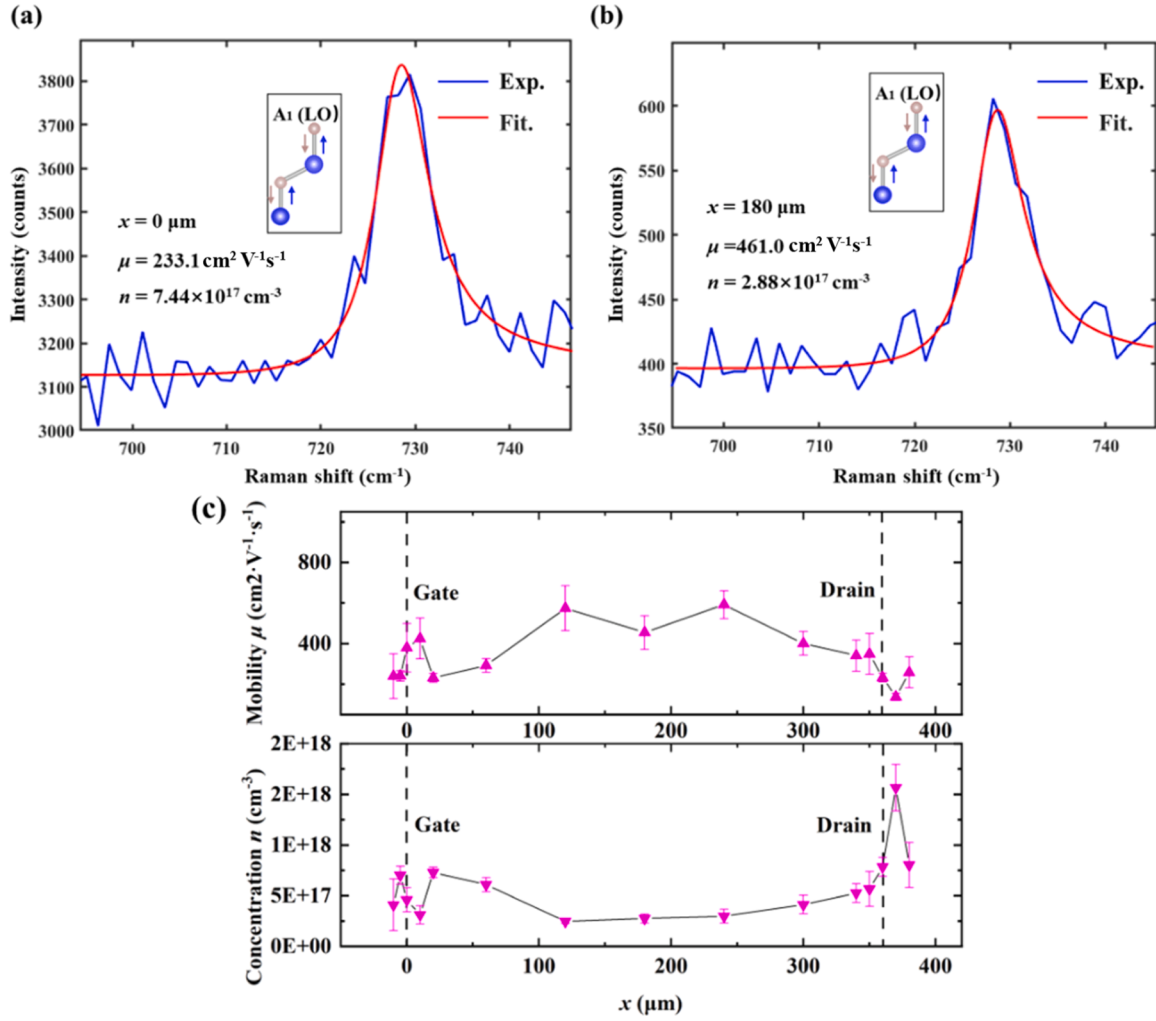


Fig. 4. Fitting of plasmon-coupled A₁(LO) mode at (a) $x = 0 \mu\text{m}$ and (b) $x = 180 \mu\text{m}$ and (c) determined carrier mobility and concentration distribution of GaN layer along x direction in non-working state.

that the carriers are scattered more at both edges. Furthermore, the pronounced variation in carrier concentration observed at $x = 370 \mu\text{m}$ suggests the influence of factors beyond residual stress, including crystal lattice quality.

After completing the Raman mode measurement, we switched to a 325nm solid-state continuous wave laser to measure the PL spectrum of GaN layer. The measurement range is 330~420nm, and the integration time is 8s. A total of 5 positions were measured from $x = -10 \mu\text{m}$ to $x = 380 \mu\text{m}$, and each position was measured 3 times. After the measurement, we used the main peak of the exponentially modified Gaussian (ExGaussian) function and three additional peaks of standard Gaussian function to fit the PL peak. Fig. 5(a) and Fig. 5(b) are the PL peak fittings for the positions $x = 0 \mu\text{m}$ and $x = -10 \mu\text{m}$, respectively. From top to bottom, each figure is the residual after subtracting the experimental curve from the fitting curve, the comparison between the experimental curve and the fitting curve, and the composite peak obtained by deconvolution. The processing effect of the composite peak is good. Further comparing the PL spectra of each position point, we found that the PL peaks in the middle area of the device (such as the position point $x = 0 \mu\text{m}$) are generally smooth, the peak shape is relatively symmetrical, and the peak width is narrow (such as the position point $x = 0 \mu\text{m}$). The PL peaks at both ends of the device (such as position $x = 0 \mu\text{m}$) are relatively rough, asymmetrical, and wide, which reflects that the materials at both ends of the device have more material defects, and also explains the degradation of carrier mobility at both edge areas

in Fig. 4.

As can be seen from Fig. 5(c), the bandgap of the GaN layer shows obvious distribution differences in the x direction. The bandgap at the edge regions on both sides is significantly smaller than that at the middle region, with a minimum of 3.402 eV and a maximum of 3.410 eV. The bandgap width in the middle region changes slightly, and the overall difference is not large. This phenomenon indicates that the regions on both sides of the GaN layer may be affected by the residual stress and preparation process, resulting in slightly poor thermal stability and slightly lower breakdown electric field strength. On the contrary, the middle region may have higher reliability in high power or frequency devices.

Table 1 compares the spatial resolution and measurement uncertainty of the Raman-PL method with other techniques for measuring multi-physics parameters. The accuracy of the Raman-PL method is primarily limited by optical scattering, laser power dependence, and local heating effects. Although its accuracy is lower than that of contact-based methods such as Hall measurements, the Raman-PL approach offers distinct advantages, including micron-scale spatial resolution and the capability for non-destructive, in-situ characterization of devices with nanostructures. Consequently, it is irreplaceable in specific applications such as analyzing local carrier distributions, interfacial transport properties, and microscale heterogeneity.

The integrated Raman-PL system provides a pathway toward comprehensive analysis of individual electronic devices. Variations in

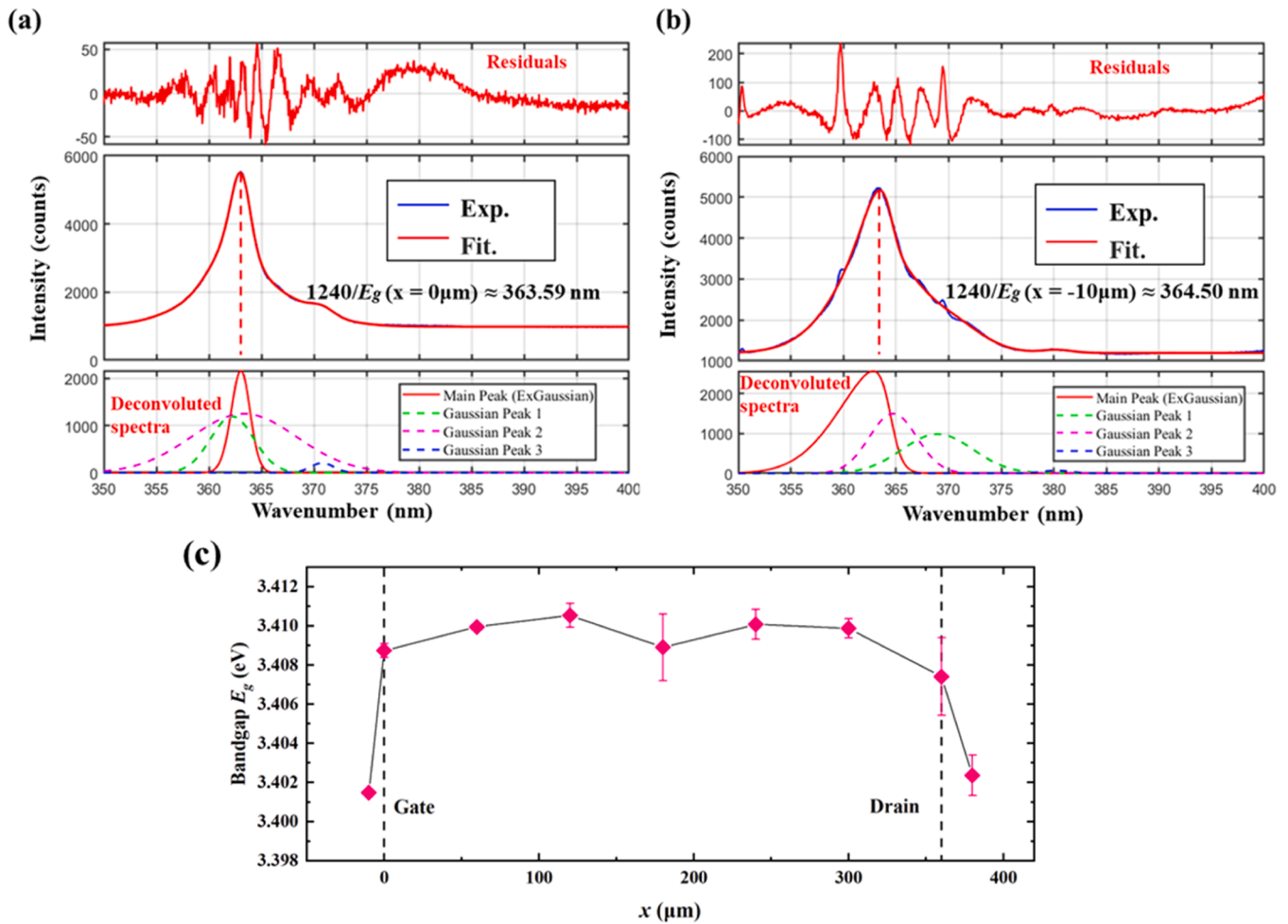


Fig. 5. Multiple peaks fitting of GaN PL spectra curve at (a) $x = 0 \mu\text{m}$ and (b) $x = -10 \mu\text{m}$ and (c) determined bandgap distribution along x direction in non-working state.

Table 1

Comparison of spatial resolution and measurement uncertainty between Raman-PL and other multi-parameter measurement techniques.

Method	Raman-PL [37, 48]	XRD [49]	Hall effect [50]	Electrical characterization [51]	UV [52]	IR [53]
Spatial resolution	$0.5\mu\text{m} \sim 2\mu\text{m}$	$1\mu\text{m} \sim 10\mu\text{m}$	Sensor-dependent	Global Information	$0.5\mu\text{m} \sim 10\mu\text{m}$	$10\mu\text{m} \sim 100\mu\text{m}$
Detection depth	$50\text{nm} \sim 2\mu\text{m}$	$5\mu\text{m} \sim 50\mu\text{m}$	Total thickness of the sample	Total thickness of the sample	$10\text{nm} \sim 100\text{nm}$	$10\mu\text{m} \sim 100\mu\text{m}$
Uncertainty in measuring carrier mobility	5%~20%	-	1%~5%	1%~5%	-	-
Uncertainty in measuring carrier concentration	~20%	-	2%~10%	2%~10%	-	-
Uncertainty in measuring bandgap	$\pm 0.01\text{eV} \sim 0.1\text{eV}$	-	-	-	$\pm 0.01\text{eV} \sim 0.1\text{eV}$	-
Uncertainty in measuring stress	$\pm 0.01\text{GPa} \sim 0.1\text{GPa}$	$\pm 0.01\text{GPa} \sim 0.1\text{GPa}$	-	-	-	-
Uncertainty in measuring temperature	$\pm 0.5\text{K} \sim 5\text{K}$	-	-	-	-	$\pm 0.5\text{K} \sim 8\text{K}$

structural design, doping, and other process parameters across different devices limit the use of a universal model, making it essential to perform comprehensive characterization on a single device within a unified, non-invasive platform. Moreover, unlike electrical measurement techniques, this method eliminates the need for external sensors, thus avoiding the perturbations that contact sensors can impose on the device's intrinsic performance, which is a notable advantage. And it can be adapted into a dual-wavelength flash Raman system for thermophysical analysis or combined with current-based techniques to investigate

carrier trap states. Although further reduction in measurement uncertainty remains necessary, this platform still offers valuable guidance for future device design by enabling simultaneous, non-contact, and non-destructive assessment of multiple parameters on the same device without system reconfiguration. This capability creates an opportunity to establish coupled multi-physics models that account for interactions among thermal, mechanical, and electrical properties. The advantages of this approach in failure modeling and analysis will be further demonstrated in the following section.

4. Failure analysis

In order to observe the actual performance of the device during operation and to perform failure analysis, we used a regulated direct-current (DC) source (RIGOL DP821A) to load the device with a voltage of $V_{gs} = -2.5V$ and $V_{ds} = 12V$, ensuring its activation and normal operation. Based on the two-step decoupling method of temperature rise and thermal stress that we proposed earlier, we simultaneously measured the temperature rise and thermal stress distribution of the GaN layer and the SiC layer, as shown in Fig. 6(a). It is crucial to calibrate the reference point for temperature rise and thermal stress measurements at $V_{gs} = -5V$ and $V_{ds} = 12V$ to preclude any influence from electrical stress [36,37]. Under this measurement condition, the temperature of both the GaN and SiC layers in the central region of the device is higher than in the side regions, and the maximum temperature rise of the GaN layer is 189.2 K, and the maximum temperature rise of the SiC layer is 175.0 K, which reflect a significant interface thermal resistance between GaN and SiC layers. The thermal stress on the GaN layer as a whole is compressive stress (maximum value is -0.05GPa, minimum value is -0.29GPa), while the thermal stress on the SiC layer as a whole is tensile stress (maximum value is 0.31GPa, minimum value is 0.20GPa). Notably, the GaN layer at both side edges experiences a dominant tensile thermal stress. Due to the poor heat dissipation performance of the printed circuit board (PCB) board on which the device is installed, the device is damaged after further voltage loading, and the failure situation is shown in Fig. 6(b). Interestingly, the most severe damage is localized at the channel edges, contrary to the central region where the temperature is highest at low drain-source voltage. This finding is not an isolated instance (as shown in Supporting Information) and is, in fact, a well-documented phenomenon in the literature, confirming its non-random nature [20].

A comparison between Fig. 6(a) and Fig. 3(b) reveals distinct variation patterns in the residual stress and thermal stress distributions. Specifically, the channel edges are under relative compression in the residual stress profile, whereas they experience relative tension under thermal stress. As the operating voltage increases, the influence of thermal stress becomes more pronounced, potentially leading to a higher propensity for failure at the edges. As shown in Eq. (8), the parameter $Q_d = \epsilon_{ox}^2 \mu / \lambda n_s$, which is governing heat generation and dissipation, is itself a function of the constraint factor f_p and directly affected by thermal stress. Thus, it can be hypothesized that the prominent edge tensile stress correlates with failure by altering hot-spot distribution via stress-induced changes in carrier mobility and

concentration.

To investigate this hypothesis, the spatial distributions of residual stress, carrier mobility, and carrier concentration on the same channel under a non-operating state were summarized in Figs. 7(a) and 7(b). The results demonstrate a clear correlation: carrier mobility increases with tensile stress, while carrier concentration decreases. The mobility variation approximates a linear relationship, whereas the concentration change deviates from linearity. Further analysis of the bandgap reveals a strong linear correlation with the stress (as shown in Fig. 7(c)), and the relationship between carrier concentration and bandgap can be expressed by the following equation:

$$n \sim T^{2/3} \exp\left(-\frac{E_g}{2k_B T}\right) \quad (19)$$

Where, k_B is the Boltzmann constant. It shows that an increase in the bandgap will cause a decrease in carrier concentration, thereby leading to increased heat generation at the tensioned location. This finding, which provides a physical basis for the observed carrier behavior, is also consistent with previous reports in the literature [54].

Substituting the above equation into the definition of Q_d , it can be

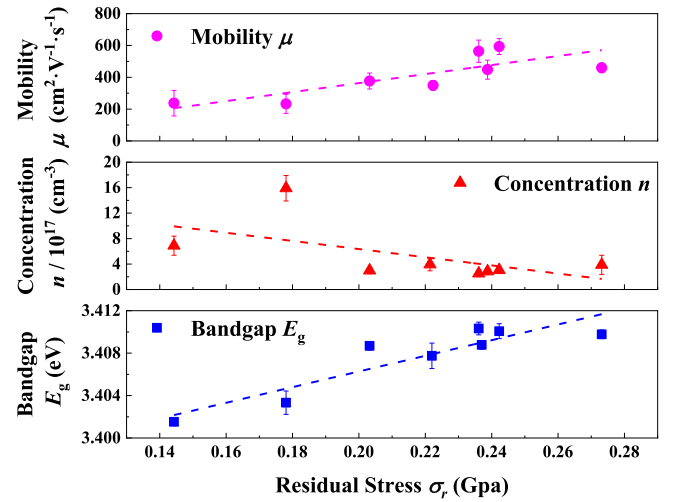


Fig. 7. The effects of residual stress on (a) carrier mobility, (b) carrier concentration, and (c) bandgap.

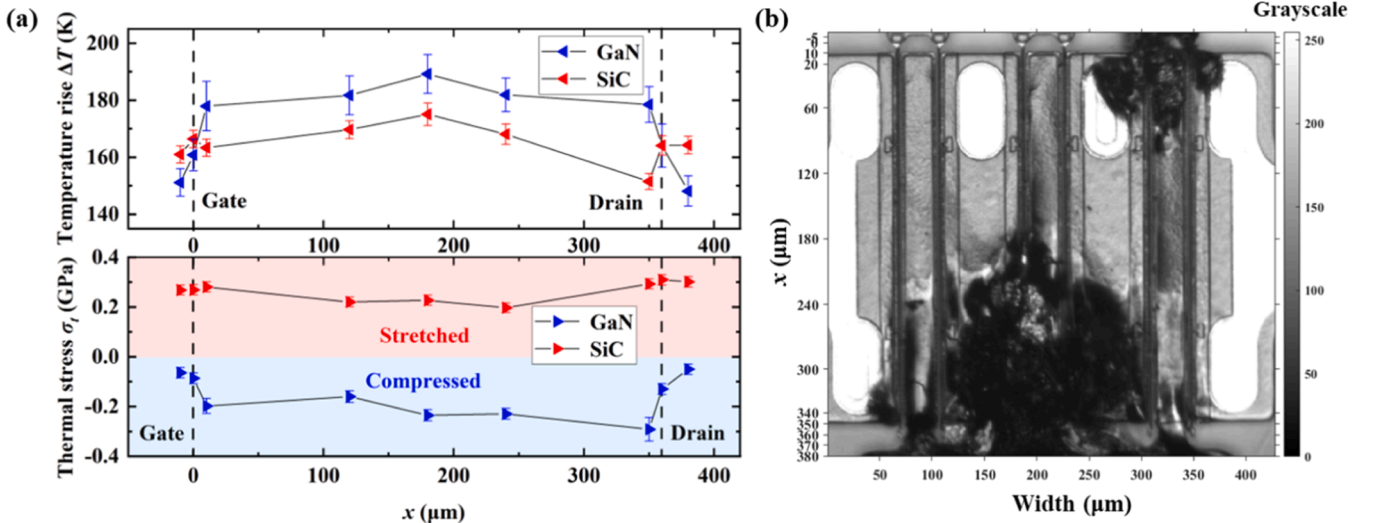


Fig. 6. (a) Temperature rise and thermal stress distribution of GaN and SiC layers ($V_{gs} = -2.5V$, $V_{ds} = 12V$) (b) Image of failure GaN HEMT after loading excessive drain-source voltage.

obtained that

$$\delta Q_d \sim \frac{k_3 \sigma}{\exp(-k_4 \sigma)} \quad (20)$$

Where, δQ_d is the relative variation of Q_d , k_3 and k_4 are constants related to the intrinsic material properties. As illustrated in Fig. 8(a), an increase in temperature rise leads to higher thermal stress, which in turn intensifies the relative tensile stress. This elevated tensile stress causes an increase in carrier mobility and a decrease in carrier concentration, resulting in a significant rise in heat generation and exacerbating local overheating. Concurrently, the material's thermal conductivity under tensile stress deteriorates substantially [14]. The combination of increased heat generation and degraded heat dissipation creates a vicious cycle, ultimately leading to the thermal burnout at the edge.

In order to further explore the influence of the significant change of material thermal properties caused by stress on the self-heating effect of the device, the electrothermal coupling of the semiconductor device was simulated. A two-dimensional single transistor model of the same size was established using the same dimensions of the device used in the experiment, as shown in the Fig. 8(b). In this model, the parameters were assigned values obtained from experimental measurements, with the thermal conductivity of GaN is set to 130 W/mK [14] and the interface thermal resistance of GaN/SiC is 10 m²K/GW. For the purpose of trend analysis, the resulting temperature field was normalized. Simulation results reveal that while the hotspot is located at the channel center with low V_{ds} , it progressively shifts toward the edges with increasing voltage due to intensified thermal stress. This phenomenon demonstrates the critical role of tensile stress in driving device failure.

It should be noted that while preferential failure at the edges may be a specific issue for this type of HEMT, the underlying physical mechanism represents a universal principle in semiconductor devices: whereby relative tensile stress induces changes in the bandgap, a decrease in carrier concentration, altered carrier mobility, and reduced thermal conductivity, collectively leading to increased heat generation and impaired heat dissipation. Owing to its non-contact nature, the presented measurement system enables real-time, in-situ monitoring of devices during operation. Furthermore, the developed analytical methodology represents a versatile framework that can be extended to a wide range of semiconductor devices. This general approach enables the analysis of thermally stressed regions arising from structural constraints and the identification of potential lattice defects, thereby facilitating the investigation of failure mechanisms and providing guidance for device design optimization.

The integration measurement provides the chance to establish a multi-physics model for a given device. However, both the measurement

method and the analytical model still present limitations that require further development. For example, existing implementations have not yet incorporated in situ measurement of the dielectric constant, which is itself influenced by stress. Additionally, the current measurements exhibit relatively high uncertainty (as analyzed in the Supporting Information), and further enhancement in measurement accuracy is needed. Nevertheless, this comprehensive strategy, comprehensive characterizing multi-physical parameters and using stress as an entry point to analyze thermal failure mechanisms, is anticipated to offer valuable guidance for future device failure assessment and optimized design.

5. Conclusions

In order to realize multi-parameter failure assessment of semiconductor devices, we integrated the two-step Raman method, plasma coupling mode analysis method and PL spectroscopy method into a unified measurement system, providing a powerful tool for comprehensive device performance analysis. Using this system, we simultaneously characterized residual stress, structural constraint factors, carrier mobility and concentration, and bandgap distribution in non-operating devices, revealing significant parameter discontinuities at both electrode edges. Under operational conditions, we further mapped temperature rise and thermal stress distributions, observing that with increasing drain-source voltage, the edge region, where subject to concentrated tensile stress, presents a higher failure risk than the central area. Combined experimental and theoretical analysis confirms that localized tensile thermal stress is the primary factor driving variations in carrier mobility, carrier concentration, and thermal conductivity, ultimately leading to device failure at the edges. This finding provides valuable insight for optimizing device architecture and enhancing reliability.

CRedit authorship contribution statement

Yupu Li: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aoran Fan:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Xiaoyu Zhang:** Writing – review & editing. **Fengyi Li:** Writing – review & editing, Funding acquisition. **Xing Zhang:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

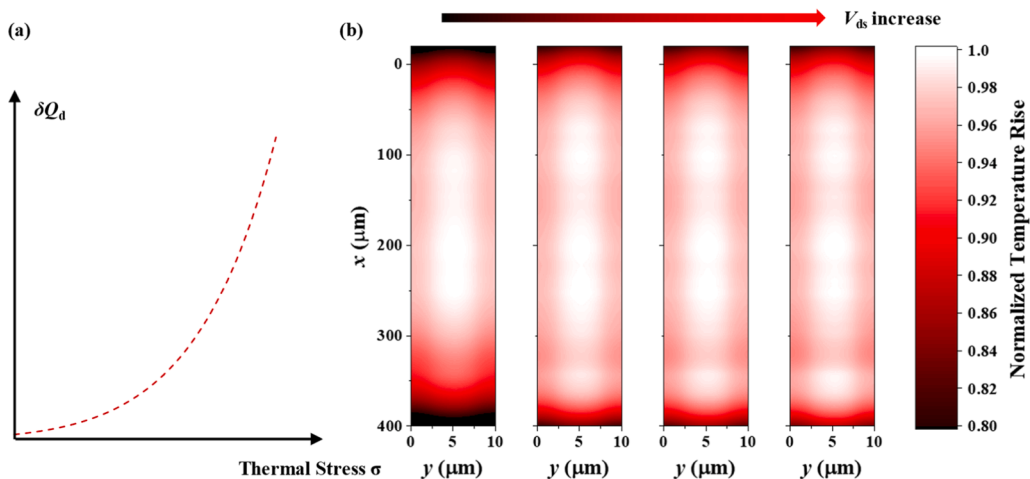


Fig. 8. (a) Schematic of δQ_d versus thermal stress σ (b) Normalized temperature rise distribution versus V_{ds} .

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijheatmasstransfer.2026.128541](https://doi.org/10.1016/j.ijheatmasstransfer.2026.128541).

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